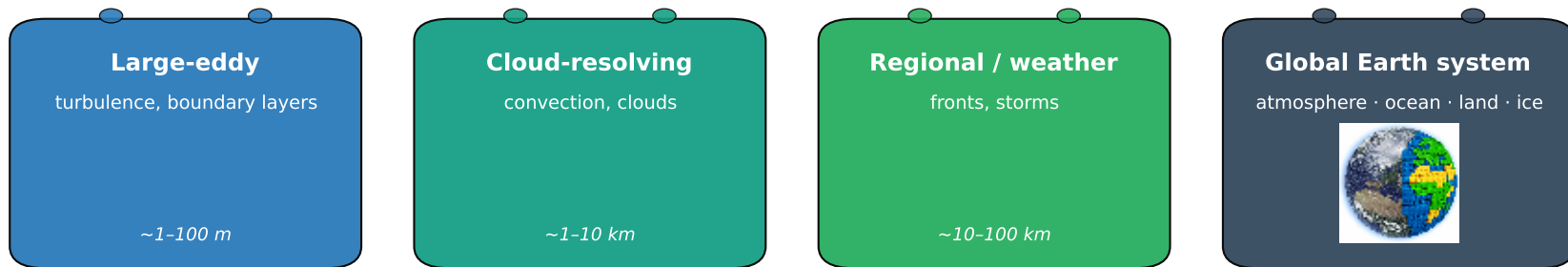


legoESM: one differentiable, composable code base across scales

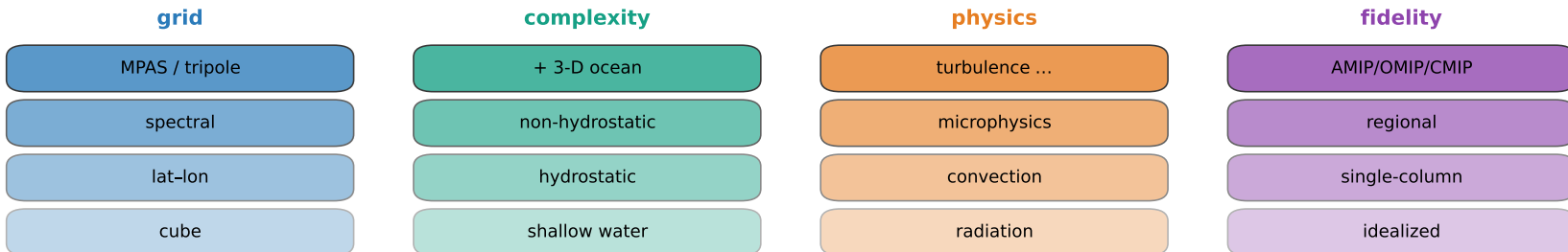
high-resolution simulations train ML components



spatial scale · weather → seasonal → centennial

learned + physical components run in the coupled global model

Composable along four independent axes — swap and stack like bricks:



end-to-end differentiable (train · calibrate · assimilate)

built and verified by AI agents under a human scientific contract